$$t_{r+1} = 11C_r (2x^2)^{11-r}\left(\frac{-3}{x}\right)^r$$

$$= 11C_r 2^{11-r} x^{2(11-r)} (-3)^r \left(\frac{1}{x}\right)^r$$

$$= 11C_r 2^{11-r} (-1)^r \cdot 3^r x^{22-3r}$$

To find the co-efficient of x^{10}, take $22-3r = 10$

$r = 4$. Put $r = 4$ in (1)

$t_5 = 11C_4 2^{11-4} 3^4 x^{10} = 11C_4 \cdot 2^7 3^4 x^{10}$

$\therefore$ Co-efficient of x^{10} is $11C_4 2^7 3^4$.

Example 2.46

Find the term independent of x in the expansion of $\left(2x + \frac{1}{3x^2}\right)^9$.

Solution

General term in the expansion of $(x + a)^n$ is $t_{r+1} = nC_r x^{n-r} a^r$

Compare $\left(2x + \frac{1}{3x^2}\right)^9$ with $(x + a)^n$

$$\therefore\ t_{r+1} = 9C_r (2x)^{9-r}\left(\frac{1}{3x^2}\right)^r$$

$$= 9C_r 2^{9-r} x^{9-r} \frac{1}{3^r} x^{-2r}$$

$$t_{r+1} = 9C_r \frac{2^{9-r}}{3^r} \cdot x^{9-3r} \quad \ldots (1)$$

To get the term independent of x, equating the power of x as 0

$9 - 3r = 0$

$\therefore\ r = 3$

Put $r = 3$ in (1)

$$t_{3+1} = 9C_3 \frac{2^{9-3}}{3^5} \cdot x^0$$

$$= 9C_3 \frac{2^6}{3^5}$$

$$= 1792$$

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Exercise 2.6

1. Expand the following by using binomial theorem
   (i) $(2a - 3b)^4$ (ii) $\left(x + \frac{1}{y}\right)^7$
   (iii) $\left(x + \frac{1}{x^2}\right)^6$
2. Evaluate the following using binomial theorem:
   (i) $(101)^4$ (ii) $(999)^5$
3. Find the 5th term in the expansion of $(x-2y)^{13}$.
4. Find the middle terms in the expansion of
   (i) $\left(x + \frac{1}{x}\right)^{11}$ (ii) $\left(3x + \frac{x^2}{2}\right)^8$
   (iii) $\left(2x^2 - \frac{3}{x^3}\right)^{10}$
5. Find the term independent of x in the expansion of
   (i) $\left(x^2 - \frac{2}{3x}\right)^9$ (ii) $\left(x - \frac{2}{x^2}\right)^{15}$
   (iii) $\left(2x^2 + \frac{1}{x}\right)^{12}$
6. Prove that the term independent of x in the expansion of $\left(x + \frac{1}{x}\right)^{2n}$ is $\frac{1 \cdot 3 \cdot 5 \ldots (2n-1) 2^n}{n!}$.
7. Show that the middle term in the expansion of $(1 + x)^{2n}$ is $\frac{1.3.5 \ldots (2n-1) 2^n x^n}{n!}$

Exercise 2.7

Choose the correct answer

1. If $nC_3 = nC_2$ then the value of nc_4 is
   (a) 2 (b) 3 (c) 4 (d) 5
2. The value of n, when $np_2 = 20$ is
   (a) 3 (b) 6 (c) 5 (d) 4

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